Multisource Feature Embedding and Interaction Fusion Network for Coastal Wetland Classification With Hyperspectral and LiDAR Data

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***Abstract*— With the development of Earth observation technology, hyperspectral image (HSI) and light detection and ranging (LiDAR) data collaborative monitoring has shown great potential in the ecological protection and restoration of coastal wetlands. However, due to the different working principles adopted by the HSI sensor and the LiDAR sensor, the data obtained by them have different distribution characteristics. The distribution difference limits the fusion of HSI and LiDAR data, bringing a great challenge for coastal wetland classification. To tackle this problem, a multisource feature embedding and interaction fusion network (MsFE-IFN) is proposed for coastal wetland classification. First, the HSI and LiDAR data are embedded in the same feature space, where the feature distribution of multisource remote sensing is aligned to alleviate data distribution differences. Second, the aligned HSI and LiDAR features interact information in channels and pixels, which is able to establish the relationship of spectral, elevation, and geospatial. Third, the HSI and LiDAR features are sent into the feature fusion network, in which the low-frequency residual is retained to enrich intraclass features. Finally, the fused feature is applied for final class prediction. Experiments conducted on three coastal wetland HSI-LiDAR datasets created by ourselves demonstrate the superiority of the proposed MsFE-IFN for coastal wetland classification. The codes will be available from the website: https://github.com/bigshotg/IEEE_TGRS_MsFE-IFN.**

***Index Terms*— Coastal wetland classification, distribution difference, feature fusion, hyperspectral image (HSI), light detection and ranging (LiDAR).**

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I. Introduction

WETLANDS are one of the three global ecosystems and are known as the kidney of the Earth. Coastal wetlands are located at the intersection of land and sea, playing an extremely important role in ecological protection, sea reclamation, and coastal protection [1], [2]. However, under the impact of human activities and natural disasters, coastal wetlands are facing the problem of ecological function degradation and biodiversity decrease [3]. Therefore, it is pressing to protect and restore coastal wetland ecology. It should be noted that the coastal wetland land cover monitoring and classification is the basis of ecological conservation and restoration.

One traditional method for coastal wetland classification is holding field survey, but this is laborious and expensive. With the development of Earth observation technology, a large number of remote sensing data bring convenience to the coastal wetland classification [4], [5]. Hyperspectral images (HSIs) have a continuous wide spectral range and a high resolution spectral response for different objects [6], [7], [8], [9], which have been widely employed in wetland classification [10]. Xie et al. [11] utilize the UAV HSIs and propose a multilayer global spectral–spatial attention network for coastal wetland classification. In [12], a probabilistic collaborative representation-based method is proposed for wetland classification, which considers the neighborhood information. To deal with the spectral aliasing of vegetation, a tiny-scene embedding approach is proposed for coastal wetland classification, which fuse the scene representation and attention mechanism [13]. In [14], an HSI transformer in transformer method is developed for coastal wetland mapping by using the ZiYuan1-02D (ZY1-02D) HSI sensor of China.

Although the above classification methods have achieved satisfactory performance in wetland classification, the vegetation of coastal wetland has the characteristics of multiecotype and multigrowth state, which increases the difficulty of coastal wetland classification. In addition, the passive operation mode of HSI remote sensing is easily affected by the weather environment, leading to the phenomenon of inconsistency between spectrum and objects [15], [16]. Fortunately, light detection and ranging (LiDAR) can actively detect ground objects regardless of weather [17], [18], [19]. The elevation information provided by LiDAR is an effective complement

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to HSI data, especially in the distinction between arbors and shrubs. Therefore, HSI and LiDAR collaborative classification is a trend of coastal wetland classification [3]. A variety of methods have been proposed toward the HSI and LiDAR classification [20], [21].

In recent years, with the rapid development of deep learning (DL) in information fusion, the DL-based approach has attracted wide attention on HSI and LiDAR classification [22], [23]. There are two patterns for the HSI and LiDAR classification. One is register HSI and LiDAR to generate HSI point cloud data [24], and the other is to fuse the HSI and LiDAR raster data at the feature or decision level [25], [26]. The former is often used for object detection or 3-D classification. For instance, Du and Zare [27] propose a multiple instance multiresolution fusion framework, which fuse the HSI and LiDAR point for scene understanding. The latter is more suitable for 2-D pixel-level classification. Considering that the coastal wetlands classification is similar to a 2-D mapping task, the method based on raster data fusion is more suitable for wetland classification.

Specifically, most of these methods focus on the extraction and fusion of multisource features [28], [29], [30]. In particular, convolutional neural networks (CNNs) have become one of the most effective model for HSI and LiDAR classification owing to its outstanding fine-grained feature extraction capability and the weight sharing strategy of convolutional kernel [31], [32], [33]. Specifically, Zhang et al. [34] propose an unsupervised feature extraction framework for HSI and LiDAR classification, which utilizes the encoder–decoder structure. In this method, a translation function between HSI and LiDAR is established to fuse the multisource remote sensing feature. Similarly, Hang et al. [35] propose a coupled CNN, using two weight sharing CNNs to extract HSI and LiDAR feature. In order to improve the performance of classification, both feature-level fusion strategy and decision-level fusion strategy are applied in its fusion model. In addition, considering the influence of remote sensing feature channels weight on classification results, the attention mechanism has been introduced in DL-based method for HSI and LiDAR feature fusion and classification. For example, Li et al. [36] develop a dual-channel CNN, which utilize the same spatial attention for HSI and LiDAR feature extraction. The classification performance verifies the effectiveness of attention mechanism. In [37], to deal with the limitations in fusing HSI and LiDAR features in a context-awareness way, a multiscale feature learning is proposed for HSI and LiDAR classification. In this method, a multilevel self-attention module and an attentional feature fusion module (FFM) are developed for feature extraction and fusion. However, because the convolution kernel of CNN has a fixed receptive field, the above CNN-based methods are unable to capture the long-range or global information. To address the problem, the graph-based method and the vision transformer (ViT)-based method have been applied for classification. Xiu et al. [38] develop a multisource attention network with discriminative graph and informative entities, which establish the topological relationship between all pixels. Zhao et al. [39] fuse a hierarchical CNN and a transformer network, which can capture the multisource heterogeneous information and improve the joint classification performance. Nevertheless, the above methods do not consider the interaction between multisource features. The landscape of coastal wetland is complex, and it is not suitable for the coastal wetland classification task that extracting feature from HSI and LiDAR without considering their interactive information.

Recently, some methods focus on the multisource feature interaction in addition to the feature extraction and fusion. The feature interaction can fully extract the relationship between multisource features and obtain more discriminative fused features. Dong et al. [40] propose a multisource aligning joint contextual representation model to fully exploit correlation of multisource data, which can interact spectral–spatial–elevation information during end-to-end training. Besides, the contrastive learning has also been applied to feature interactions. For instance, in [41], a fusion method of collaborative contrastive learning is proposed to achieve the coordination feature interaction and matching between HSI and LiDAR data and has an excellent performance on small sample classification tasks. Wang et al. [42] develop a robust self-supervised contrastive learning network, which is based on the nearest neighbor method. In this method, the multisource feature semantic alignments can be captured accurately, and the multi-order feature interactions between the HSI and LiDAR feature are exploited. However, although the above feature interaction-based approach has competitive classification performance, these methods ignore the distribution differences in HSI and LiDAR data caused by sensor modality discrepancy.

Due to the different working principles and measurement method adopted by the HSI sensor and LiDAR sensor, the data obtained by them have different distribution characteristics. In general, this difference needs to be considered in data processing and analysis, because it can affect the performance of HSI and LiDAR joint classification. But, the above joint classification methods directly input multisource remote sensing data into the feature extraction network and do not consider the distribution variations of HSI and LiDAR data. Moreover, owing to the complex and diverse landscape of coastal wetland, it is more sensitive to the variations of data distribution. Accordingly, facing the data distribution differences, designing an effective feature extraction, interaction, and fusion network becomes a challenge for coastal wetland classification.

To address this issue, a multisource feature embedding and interaction fusion network (MsFE-IFN) is proposed for coastal wetland classification. First, the proposed MsFE-IFN is able to expand the dimension of LiDAR data, aligning the HSI and expanded LiDAR to reduce the distribution differences. Second, the aligned HSI and LiDAR features establish the relationship of spectral and elevation by channel interact and then introduce the geospatial information by spatial interact. Third, a two-stage fusion network is applied to fuse HSI and LiDAR features for final classification. Three real-world datasets collected in Yellow River Delta coastal wetland are used for experiments, showing the competitive performance with other methods. The main contributions of MsFE-IFN can be summarized as follows.

1) An MsFE-IFN is explored for coastal wetland classification. Different from the existing works, the MsFE-IFN

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is able to relief the influence of sensor modality discrepancy. In addition, the approach of interaction fusion of features is employed to capture the discriminative features, which is more suitable for the complex landscape classification in coastal wetland.

2) A diverse features expansion and distribution alignment module (DFE-DAM) is presented to alleviate the data distribution differences of HSI and LiDAR. In the module, the orthogonal loss is designed to enforce the expanded features to learn more informative representations, and the joint Kullback–Leibler divergence loss (JKL loss) is introduced to control the HSI and LiDAR features alignment.
3) A cross-modality feature interaction module (CMFIM) is proposed to perform the information interaction of HSI and LiDAR feature. The channelwise interaction mechanism is applied to establish the interrelation of spectral and elevation. The spatialwise interaction mechanism is deployed to capture the geospatial relationship, which is based on the observation target consistency of HSI and LiDAR.
4) To fuse the HSI and LiDAR feature, a two-stage fusion module is developed for feature fusion. In the first stage, the multihead cross-attention mechanism is adopted to carry out the sequence-to-sequence fusion. In the second stage, the fusion module introduces the low-frequency residual to enrich intraclass features, which increases the discriminability of fusion features.

The rest of this article is organized as follows. Section II expounds the study area and the experimental data acquisition. Section III presents the details of the proposed method MsFE-IFN. Section IV discusses the experimental results and analyzes the effectiveness of the proposed method. The conclusion of this article is shown in Section V.

II. Materials

A. Study Area and Data Acquisition

1) Yellow River Delta: The study area is located in the Yellow River Delta National Nature Reserve, Shandong, China. Yellow River Delta is an alluvial plain formed by sediment deposited by the Yellow River in the Bohai Depression, which is a special coastal wetland in the world. The dominant vegetation species in Yellow River Delta coastal wetland are Suaeda salsa, Tamarix chinensis, Salix matsudana, Phragmites australis, and so on. Besides, there are also large areas of artificial Robinia pseudoacacia in the nature reserve. We selected three typical areas as the study area, and two field surveys are conducted in September 2022 and June 2023 (Fig. 1). The three typical areas are located in the north shore of the Yellow River, and the three dominant species of Tamarix forest, Robinia acacia forest, and Saliz matsudana are studied, respectively. The area of the three areas is 15.86, 40.25, and 36.57 ha, respectively.

2) UAV-Based Data: The UAV is utilized to acquire the HSI and LiDAR data of three typical areas.

a) Hyperspectral data: The HSI data are collected by the frame camera Cubert Utilities FIREFLEYE S185, which

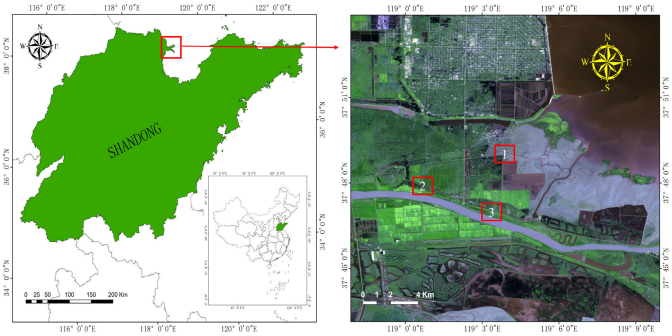


Fig. 1. Overview of the study area.

is carried by the six-rotor UAV platform DJI M600 Pro. The flight route is predetermined with flight altitude about 100 or 150 m, and the camera is set to take pictures at equidistant intervals automatically. The size of obtained single HSI image is 1000 × 1000 pixels, and the spatial resolution is about 0.4 or 0.6 cm. The HSI data contain 126 spectral bands with a spectral resolution of 4 nm, ranging from 450 to 950 nm. All flights are conducted on sunny and windless weather. The HSI images are collected between 10:00 and 15:30.

b) LiDAR data: The LiDAR data are obtained by the D-LiDAR 2000 camera, which is carried by the four-rotor UAV platform FEIMA D2000s. The flight route is also predetermined with flight altitude about 50 or 120 m, and the scanning angle is about 70°. The laser wavelength is 905 nm, and the number of echo is three. The point frequency is 240 kHz, and the average density of obtained point clouds is 220 or 94 points/m^2. All flights are conducted simultaneously with the HSI-UAV.

The data acquired from UAV should be preprocessed before using. First, the original HSI data are conducted the radiometric calibration and geometric correction. Second, each HSI image is aligned to get an HSI image of the entire flight area. Third, the original point clouds are conducted the point resolving, strip adjustment, and raster conversion to get the LiDAR image of the flight area. In general, the small dominant vegetation species in coastal wetlands are Suaeda salsa and Tamarix chinensis, and their crown breadth ranges from 0.2 to 1 m. Too high spatial resolution makes pixels lose the overall structure and distribution of vegetation, which is not conducive to pixel classification. Considering that submeter pixel is more suitable for wetland land cover classification, the spatial resolution of HSI and LiDAR is resampled to 0.5 m. Next, the HSI and LiDAR data are projected into the projected coordinate system WGS 1984 UTM Zone 50N, aligned pixel by pixel to acquire the multisource dataset. Finally, combined with field data and UAV-based RGB images, the multisource remote sensing data are interpreted to get wetland classification datasets.

B. Datasets

After data preprocessing, three coastal wetland classification datasets are obtained, corresponding to three typical areas, respectively.

1) Tamarisk Dataset: The first typical area is named Tamarisk dataset. Tamarix chinensis is a typical vegetation of

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TABLE I
NUMBERS OF TRAINING AND TEST SAMPLES FOR THE TAMARISK DATASET

No.	color	Class	Training	Test
1		Small tidal creek	200	4235
2		Phragmites australis	200	9660
3		Desiccated Spartina alterniflora	200	52284
4		Abandoned oil platform	50	150
5		Tamarix chinensis	200	33544
6		Suaeda salsa	200	24305
7		Large tidal creek	200	47660
8		TC and SS mixed	200	5448
9		Roads	200	2691
		Total	1650	179977

TABLE II
NUMBERS OF TRAINING AND TEST SAMPLES FOR THE ROBINIA PSEUDOACACIA DATASET

No.	color	Class	Training	Test
1		Phragmites australis	200	106498
2		Robinia pseudoacacia	200	93575
3		Water	200	133037
4		Salix matsudana	200	47636
5		Tamarix chinensis	200	669
6		Rice paddy	200	67237
7		Highway	200	12383
8		Abandoned farmland	200	42630
9		Firebreak ditch	200	7678
10		Dirt road	200	7402
11		Withered Robinia pseudoacacia	200	9036
		Total	2200	527781

TABLE III
NUMBERS OF TRAINING AND TEST SAMPLES FOR THE NATURAL WILLOW DATASET

No.	color	Class	Training	Test
1		Salix matsudana	200	102049
2		Senescent Phragmites australis	200	105427
3		Populus	200	12394
4		Tamarix chinensis	200	48795
5		Yellow River diversion canal	200	38615
6		Dirt road 1	200	5643
7		Dirt road 2	200	4764
8		Highway	200	13549
9		Farmland	200	3037
10		Yellow River riverbed	200	60464
11		Pool	200	15654
12		Phragmites australis	200	84758
		Total	2400	495149

ecological evolution in the Yellow River Delta coastal wetland. The Tamarisk dataset includes nine classes, i.e., Tamarix chinensis, Suaeda salsa, Tamarix chinensis and Suaeda salsa mixed (TC and SS mixed), Phragmites australis, Desiccated Spartina alterniflora, Large tidal creek, Small tidal creek, Abandoned oil platform, and Road. The area covered contains 784×1024 pixels. After removing background pixels, 179 977 samples are maintained. Table I shows the detailed information of training and test samples.

2) Robinia Pseudoacacia Dataset: The second typical area is named Robinia pseudoacacia dataset. Robinia pseudoacacia has been widely planted in the Yellow River Delta reserve, which can improve the ecological environment of coastal wetlands. Robinia pseudoacacia dataset has 1484×1246 pixels and contains 527 781 labeled samples. It contains 11 types of ground objects, i.e., Phragmites australis, Robinia pseudoacacia, Water, Salix matsudana, Tamarix chinensis, Rice paddy, Highway, Abandoned farmland, Firebreak ditch, Dirt road, and Withered Robinia pseudoacacia. The information of training and test samples is listed in Table II.

3) Natural Willow Dataset: The third typical area is named Natural willow dataset. The Yellow River carries Salix matsudana seeds from the middle reaches to the lower reaches. Salix matsudana is widely distributed in the Yellow River estuary, which is an important terrestrial vegetation in the Yellow River Delta coastal wetlands. The imagery size of Natural willow dataset is 1197×1445 pixels. After removing background pixels, Natural willow dataset has 495 149 samples. It includes 12 classes, i.e., Salix matsudana, Senescent Phragmites australis, Populus, Tamarix chinensis, Yellow River diversion canal, Dirt road 1, Dirt road 2, Highway, Farmland, Yellow River riverbed, Pool, and Phragmites australis. Table III shows the detailed information of training and test samples.

The pseudo-color image, the LiDAR raster image, and the ground-truth map of three datasets are shown in Fig. 2.

III. METHODOLOGY

In this section, we introduce the architecture and mathematical understanding of MsFE-IFN in detail. The overall framework of the proposed MsFE-IFN is depicted in Fig. 3. MsFE-IFN is composed of three modules: 1) DFE-DAM; 2) CMFIM; and 3) FFM. Detailed information on each part of MsFE-IFN is described in Sections III-A–III-C.

A. Diverse Features Expansion and Distribution Alignment Module

HSI sensors obtain spectral information in continuous spectral range by recording light radiation of different wavelengths, while LiDAR sensors emit laser pulses and measure the transmission time of the returned laser pulses to obtain the elevation information of objects. The difference of working principle and data acquisition model between HSI and LiDAR is called sensor modality discrepancy. The sensor modality

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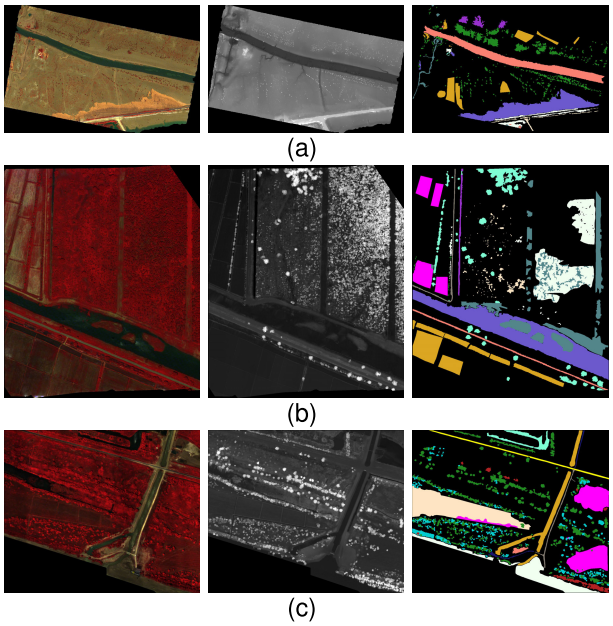


Fig. 2. Pseudo-color image, the LiDAR raster image, and the ground-truth map for (a) Tamarisk dataset, (b) Robinia pseudoacacia dataset, and (c) Natural willow dataset.

discrepancy leads to the data distribution difference between HSI and LiDAR. The data distribution difference brings many irrelevant information in feature fusion, which leads to low discriminability of obtained features. Accordingly, the problem of data distribution difference in multisource remote sensing data is the main factor affecting joint classification performance of HSI and LiDAR. To address this issue, the DFE-DAM is developed to reduce data distribution differences of HSI and LiDAR. Concretely, the DFE-DAM is divided into two stages: the LiDAR features expansion stage and the HSI-LiDAR features alignment stage.

1) LiDAR Features Expansion: There is a huge dimensional differences between HSI and LiDAR data, which inhibits the complementary capability of HSI and LiDAR. Hence, the LiDAR features expansion stage aims to generate more features embedding for LiDAR, which ensure that the LiDAR features have the same channel dimensions as the HSI features. To be specific, a dual-branch convolutional generation structure is applied for LiDAR features expansion. For each branch, three 1×1 convolutional layers $\varphi^1_{1\times1}$, $\varphi^2_{1\times1}$, and $\varphi^3_{1\times1}$ are utilized to generate LiDAR features with different dimensions. $\varphi^1_{1\times1}$, $\varphi^2_{1\times1}$, and $\varphi^3_{1\times1}$ can map the input data into different feature spaces, which is able to enrich the representation of LiDAR features. Then, the batch normalization (BN) layer F_{BN} is deployed to carry out feature scaling operation, and the rectified linear unit (ReLU) activation layer F_{ReLU} is employed to improve the nonlinear representation capability of features. Finally, the features of different spaces are concatenated, and a 1×1 convolutional block is used to fuse the different features. Let us assume that the input LiDAR data are $\boldsymbol{X}_L \in \mathbb{R}^{h\times w\times 1}$, which is the subimage split by original LiDAR. h and w are the patch size of $\boldsymbol{X}_L$. The generated features of each branch can be formulated as follows:

$$\boldsymbol{F}^i_{\mathrm{lidar}} = \phi^i_{1\times1}\Big[f^1_{\mathrm{BR}}\big(\varphi^1_{1\times1}(\boldsymbol{X}_L)\big), f^2_{\mathrm{BR}}\big(\varphi^2_{1\times1}(\boldsymbol{X}_L)\big), \times\ f^3_{\mathrm{BR}}\big(\varphi^3_{1\times1}(\boldsymbol{X}_L)\big)\Big] \tag{1}$$

where $\boldsymbol{F}^i_{\mathrm{lidar}} \in \mathbb{R}^{h\times w\times n}$ is the output features of each branch, $n = (d/2)$ is the dimensions of features, and $i \in [1, 2]$ is the branch number. $\phi^i_{1\times1}$ is the 1×1 conv–BN–ReLU block. f^n_{BR} is the function block of $F_{\mathrm{ReLU}}(F_{\mathrm{BN}}(\cdot))$.

As shown in Fig. 3, $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$ are concatenated together as the final LiDAR feature $\boldsymbol{F}_L \in \mathbb{R}^{h\times w\times d}$. d is the dimensions of $\boldsymbol{F}_L$ and equal to $2n$ numerically. $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$ are the output feature of the dual-branch convolutional generation structure, which are the different embedding representations of the input $\boldsymbol{X}_L$. The dual-branch structure aims to generate the informative feature representations, while unconstrained concatenation of the two features will inevitably result in information redundancy. Therefore, the first property is applied to constrain $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$.

Property 1: The $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$ should be as diverse as possible to effectively learn the informative feature representations.

This means that $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$ are independent of each other and have mutually exclusive information representations, which can ensure that the $\boldsymbol{F}_L$ contains the most abundant representation features and the least redundant information. To achieve this goal, the orthogonal loss is designed to enforce the expansion features to learn diverse features representations. The $\boldsymbol{F}^1_{\mathrm{lidar}}$ and $\boldsymbol{F}^2_{\mathrm{lidar}}$ are orthogonal represents that they exist in two separate subspaces, which can capture elevation changes flexibly and enrich elevation feature representation. Thus, the orthogonal loss can be calculated as follows:

$$L_{\mathrm{ort}} = \sum_{i=1}^{n}\Big[\big(\boldsymbol{F}^1_{\mathrm{lidar}}\big)^{\mathrm{T}} \cdot \boldsymbol{F}^2_{\mathrm{lidar}}\Big]_i \tag{2}$$

where n is the number of subimages.

2) HSI-LiDAR Features Alignment: The HSI-LiDAR features alignment stage aims to achieve feature alignment and mitigate the data distribution difference between HSI and LiDAR. First, a 1×1 convolutional block and two 3×3 convolutional blocks are used to map the HSI from data source domain to the feature space, which make the HSI features have the same dimension as LiDAR features. Then, assuming that the output HSI features are $\boldsymbol{F}_H \in \mathbb{R}^{h\times w\times d}$, the $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ should be align to the same feature distribution. Hence, the second property is applied to constrain $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$.

Property 2: The $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ should facilitate reducing the distribution discrepancy between HSI and LiDAR.

This means that we need to pull close the distribution distances between $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. To this end, the KL loss is first adopted to constrain $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. However, KL loss is asymmetrical, and choosing either $\boldsymbol{F}_H$ or $\boldsymbol{F}_L$ as the reference distribution will lead to information loss of the other. Therefore, the JKL loss is designed to constrain $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ from two directions, which give equal weight to $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. The JKL loss can be calculated as follows:

$$L_{\mathrm{JKL}} = \sum_n \sum_{ij}\left[\alpha \boldsymbol{P}_{(F_H)} \log\left(\frac{\boldsymbol{P}_{(F_H)}}{\boldsymbol{Q}_{(F_L)}}\right) + \beta \boldsymbol{Q}_{(F_L)} \log\left(\frac{\boldsymbol{Q}_{(F_L)}}{\boldsymbol{P}_{(F_H)}}\right)\right] \tag{3}$$

where n is the number of subimages, (i, j) is the index of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$, and $\boldsymbol{P}_{(F_H)}$ and $\boldsymbol{Q}_{(F_L)}$ are the probability distribution

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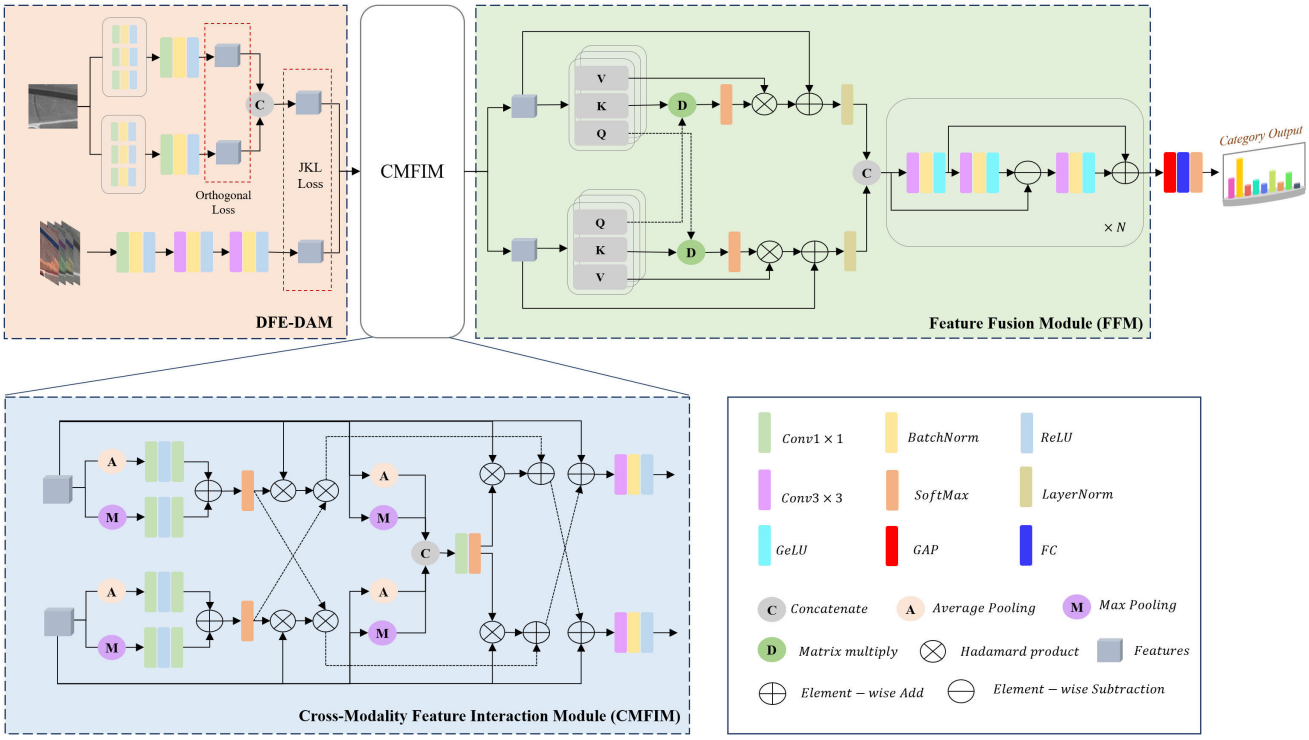


Fig. 3. Overall framework of the proposed MsFE-IFN, including DFE-DAM, CMFIM, and FFM. In particular, the orthogonal loss and JKL loss are used for LiDAR and HSI expansion, respectively. The CMFIM contains spectral and geospatial interaction, and FFM has the sequence and low-frequency fusion. Both CMFIM and FFM aim to improve the discriminability of features.

of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. α and β are the hyperparameters ($\alpha+\beta=1$), which control the weight of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. $\alpha=\beta=0.5$, giving the equal weight to $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$.

Under the constraints of properties 1 and 2, $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ have similar feature distribution and reduce the data distribution differences of HSI and LiDAR, which is more suitable for the fusion of multisource remote sensing features.

B. Cross-Modality Feature Interaction Module

As mentioned above, the information originating from HSI and LiDAR is complementary, while the complementary information can be mined by the feature interaction. The complementary information is able to explore the correlation of multisource feature, which is beneficial to capture the discriminative fusion features. Therefore, the CMFIM is proposed to interact the HSI and LiDAR features, which has two stages: 1) the channelwise feature interaction stage and 2) the spatialwise feature interaction stage. The CMFIM offers a holistic interaction, which is enabling to acquire the interrelation of spectral–elevation–geospatial information.

1) Channelwise Feature Interaction: For coastal wetland, the elevation of object is correlated with the spectrum. For instance, the vegetation of the same species, with good growth and high height, will have stronger reflectance in the near infrared band. Thus, the channelwise feature interaction stage is proposed to explore the interrelation of spectral and elevation. The $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ obtained from DFE-DAM have the same number of channels, and the channelwise interaction is able to perform the exchange of spectral and elevation information. As shown in Fig. 3, first, the global average pooling (GAP) operation and the global max pooing (GMP) operation are applied to $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ along the channel dimension to remain more information. Second, the squeeze-excitation mechanism, consisting of two 1×1 convolutional layers $\phi^{\text{sq}}_{1\times1}$ and $\phi^{\text{ex}}_{1\times1}$ and one ReLU activation layer F_{ReLU}, is utilized on the output of GAP and GMP to learn the dependencies between different channels. Then, an elementwise addition operation and a softmax function are employed to obtain two channel attention vectors $\boldsymbol{A}^C_H \in \mathbb{R}^{1\times d}$ and $\boldsymbol{A}^C_L \in \mathbb{R}^{1\times d}$. The mathematical understanding for the attention vectors is as follows:

$$\boldsymbol{A}^C_i = \text{softmax}([\phi_{\text{se}}(\text{GAP}(\boldsymbol{F}_i)) + \phi_{\text{se}}(\text{GMP}(\boldsymbol{F}_i))]) \quad (4)$$

where $\boldsymbol{A}^C_i \in [\boldsymbol{A}^C_H, \boldsymbol{A}^C_L]$ and $\boldsymbol{F}_i \in [\boldsymbol{F}_H, \boldsymbol{F}_L]$. ϕ_{se} is the squeeze–excitation mechanism block $\varphi^{\text{ex}}_{1\times1}(F_{\text{ReLU}}(\varphi^{\text{sq}}_{1\times1}(\cdot)))$.

Next, the $\boldsymbol{A}^C_H$ and $\boldsymbol{A}^C_L$ are going to act on $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$, respectively. It can adaptively assign weights to different spectral channels and elevation channels, improving the characterize ability of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$. Finally, with channel attention exchange, the $\boldsymbol{A}^C_H$ and $\boldsymbol{A}^C_L$ act on elevation features and spectral features, respectively. By this way, the spectral features interact with the elevation features, establishing the relationship of spectral–elevation. It can be defined as follows:

$$\begin{aligned}\boldsymbol{F}^C_H &= \left(\boldsymbol{F}_H \odot \boldsymbol{A}^C_H\right) \odot \boldsymbol{A}^C_L \\ \boldsymbol{F}^C_L &= \left(\boldsymbol{F}_L \odot \boldsymbol{A}^C_L\right) \odot \boldsymbol{A}^C_H\end{aligned} \quad (5)$$

where $\boldsymbol{F}^C_H, \boldsymbol{F}^C_L \in \mathbb{R}^{h\times w\times d}$ are the HSI and LiDAR features with spectral–elevation interactive information. $\odot$ is the channelwise multiplication.

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2) Spatialwise Feature Interaction: As the aforementioned channelwise feature interaction stage concentrates on exploring the spectral–elevation interactive information, a spatialwise feature interaction is further introduced for acquire the geospatial interactive information. Geospatial information provides context information about pixels, which captures the correlation between pixels and improves the feature robustness. Hence, based on the consistent observation targets of HSI and LiDAR, the spatial interaction between $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ is developed. First, the GAP and GMP are employed on $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$, obtaining four spatial feature weight maps. Different from the channelwise feature interaction stage, the four weight maps are concatenated for one maps. Then, a 1×1 convolutional layers is applied, followed with a softmax function to acquire the geospatial feature weight maps. Finally, with shared geospatial feature weights, the interaction of HSI and LiDAR geospatial information is performed. The spatialwise feature interaction stage can be specified as follows:

$$\boldsymbol{A}^S = \phi_{\text{cs}}[\text{GAP}(\boldsymbol{F}_H), \text{GMP}(\boldsymbol{F}_H), \text{GAP}(\boldsymbol{F}_L), \text{GMP}(\boldsymbol{F}_L)] \tag{6}$$

$$\boldsymbol{F}_i^S = \boldsymbol{F}_i \star \boldsymbol{A}^S \tag{7}$$

where $\boldsymbol{A}^S \in \mathbb{R}^{h\times w\times 1}$ is the geospatial feature weight and ϕ_{cs} is the block of 1×1 convolutional kernel and softmax function. $\boldsymbol{F}_i^S \in [\boldsymbol{F}_H^S, \boldsymbol{F}_L^S \in \mathbb{R}^{h\times w\times d}]$ is the HSI and LiDAR features with geospatial interactive information, and $\star$ is the spatialwise multiplication.

To acquire the interrelation of spectral–elevation–geospatial information, the information exchange of $\boldsymbol{F}_i^C$, $\boldsymbol{F}_i^S$ and $\boldsymbol{F}_H$, $\boldsymbol{F}_L$ is carried out. As shown in Fig. 3, the whole feature interaction for both $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ is formulated as follows:

$$\begin{aligned}\boldsymbol{F}_H^{CS} &= \phi_{3\times 3}\big(\boldsymbol{F}_H + \boldsymbol{F}_L^C + \boldsymbol{F}_L^S\big)\\ \boldsymbol{F}_L^{CS} &= \phi_{3\times 3}\big(\boldsymbol{F}_L + \boldsymbol{F}_H^C + \boldsymbol{F}_H^S\big)\end{aligned} \tag{8}$$

where $\phi_{3\times 3}$ is the 3×3 conv–BN–ReLU block and $\boldsymbol{F}_H^{CS}, \boldsymbol{F}_L^{CS} \in \mathbb{R}^{h\times w\times d}$ is the features after the comprehensive interaction, which will be sent into the FFM for feature fusion.

C. Feature Fusion Module

After obtaining the interactive features of HSI and LiDAR, the FFM is proposed to fuse the HSI and LiDAR features into a single feature, enhancing the feature discriminability, robustness. The FFM has two stages: 1) the sequence-to-sequence fusion stage and 2) the low-frequency residual enrichment stage. The first stage is utilized to complete the sequence-level fusion of features, and the second is applied for features merging.

1) Sequence-to-Sequence Fusion: At this stage, the multihead cross-attention mechanism is developed to perform the integration of complementarities between features $\boldsymbol{F}_H^{CS}$ and $\boldsymbol{F}_L^{CS}$, completing the sequence-to-sequence fusion. First, the input multisource remote sensing feature $\boldsymbol{F}_i^{CS} \in \mathbb{R}^{h\times w\times d}$ will be reshaped as $\boldsymbol{F}_{i_\text{re}}^{CS} \in \mathbb{R}^{hw\times d}$. Then, three independent domains $\boldsymbol{Q}_i, \boldsymbol{K}_i, \boldsymbol{V}_i \in \mathbb{R}^{hw\times d}$ are derived from $\boldsymbol{F}_{i_\text{re}}^{CS}$, which are named query $\boldsymbol{Q}_i$, key $\boldsymbol{K}_i$, and value $\boldsymbol{V}_i$, respectively. It can be calculated as follows:

$$\boldsymbol{Q}_i = \boldsymbol{F}_{i_\text{re}}^{CS}\boldsymbol{W}_i^Q, \boldsymbol{K}_i = \boldsymbol{F}_{i_\text{re}}^{CS}\boldsymbol{W}_i^K, \boldsymbol{V}_i = \boldsymbol{F}_{i_\text{re}}^{CS}\boldsymbol{W}_i^V \tag{9}$$

where $\boldsymbol{W}_i^Q, \boldsymbol{W}_i^K, \boldsymbol{W}_i^V \in \mathbb{R}^{d\times c}$ are the learnable transformation matrices. $\boldsymbol{Q}_i$, $\boldsymbol{K}_i$, and $\boldsymbol{V}_i$ are divided into n heads embedding, and the dimension of each head is (c/n).

Next, the multihead cross-attention feature maps $\boldsymbol{A}_i^{\text{ca}} \in \mathbb{R}^{(c/n)\times(c/n)}$ are obtained by multiplying the query $\boldsymbol{Q}_i$ and the key $\boldsymbol{K}_i$ from the other modality path. The cross attention is able to find the most valuable information about this modality in another modality and fusing the complementary information. The formula of multihead cross-attention feature map is as follows:

$$\boldsymbol{A}_i^{\text{ca}} = \text{softmax}\big(\boldsymbol{Q}_i^{\text{T}}\boldsymbol{K}_i\big) \tag{10}$$

where $\boldsymbol{A}_i^{\text{ca}} = [\boldsymbol{A}_H^{\text{ca}}, \boldsymbol{A}_L^{\text{ca}}]$ and $\boldsymbol{Q}_i^{\text{T}}$ is the transposed matrix of $\boldsymbol{Q}_i$. Then, the cross attention of each head is calculated, and the output features of each head are fused along channelwise. Finally, a residual connection and a layer normalization layer $\boldsymbol{F}_{\text{LN}}$ are applied to get the sequence-level fusion feature

$$\begin{aligned}\boldsymbol{F}_H^{\text{se}} &= \boldsymbol{F}_{\text{LN}}\big(\text{proj}\big(\boldsymbol{V}_H\boldsymbol{A}_H^{\text{ca}}\big)\big) + \boldsymbol{F}_H^{CS}\\ \boldsymbol{F}_L^{\text{se}} &= \boldsymbol{F}_{\text{LN}}\big(\text{proj}\big(\boldsymbol{V}_L\boldsymbol{A}_L^{\text{ca}}\big)\big) + \boldsymbol{F}_L^{CS}\end{aligned} \tag{11}$$

where $\text{proj}(\cdot)$ is the linear projection, which is utilized to fuse multihead features. The $\boldsymbol{F}_H^{\text{se}}, \boldsymbol{F}_L^{\text{se}}$ are the sequence-level fusion feature, which will be concatenated and sent to next fusion stage. The final output features $\boldsymbol{F}^{\text{se}}$ of this stage can be calculated as follows:

$$\boldsymbol{F}^{\text{se}} = \big[\boldsymbol{F}_H^{\text{se}}, \boldsymbol{F}_L^{\text{se}}\big]. \tag{12}$$

2) Low-Frequency Residual Enrichment: In the second stage of FFM, $\boldsymbol{F}_H^{\text{se}}, \boldsymbol{F}_L^{\text{se}}$ will be fused as features with highly discriminative for classification, which use a low-frequency residual enrichment mechanism. To be specific, the second stage of FFM has N fusion units, and each fusion units has three 3×3 conv–BN–ReLU blocks. The purpose of fusion unit is to downsample features to get fusion features. Let us assume that the outputs of each 3×3 conv–BN–ReLU block are $\boldsymbol{g}_s, \boldsymbol{g}_e$, and $\boldsymbol{g}_{\text{lf}}$, respectively. The formula of each fusion units is as follows:

$$\begin{aligned}\boldsymbol{g}_s &= \phi_{3\times 3}^{i_1}\big(\boldsymbol{F}_{\text{in}}^i\big)\\ \boldsymbol{g}_e &= \phi_{3\times 3}^{i_2}\big(\boldsymbol{g}_s\big)\\ \boldsymbol{g}_{\text{lf}} &= \phi_{3\times 3}^{i_3}\big(\boldsymbol{F}_{\text{in}}^i - \boldsymbol{g}_e\big)\end{aligned} \tag{13}$$

where the $\boldsymbol{F}_{\text{in}}^i$ is the inputs of the ith fusion units, and $\boldsymbol{F}_{\text{in}}^1 = \boldsymbol{F}^{\text{se}}$. $\phi_{3\times 3}^{i_1}$, $\phi_{3\times 3}^{i_2}$, and $\phi_{3\times 3}^{i_3}$ are the 3×3 conv–BN–ReLU blocks. The first block is deployed to downsample features $\boldsymbol{F}_{\text{in}}^i$ to $\boldsymbol{g}_s$. However, in the process of convolution, more high-frequency information, such as texture, will be obtained, while low-frequency information will be lost. The low-frequency information contains the invariant feature of intraclass. Therefore, the low-frequency residual enrichment mechanism is proposed to retain the low-frequency information. The second block is utilized to upsample $\boldsymbol{g}_s$ to $\boldsymbol{g}_e$, which has the same size of $\boldsymbol{F}_{\text{in}}^i$ and can be used to calculate the low-frequency residual $\boldsymbol{g}_{\text{lf}}$. The third blocks downsample the low-frequency residual to $\boldsymbol{g}_{\text{lf}}$, which has the same size of $\boldsymbol{g}_s$. Finally, adding $\boldsymbol{g}_s$ and $\boldsymbol{g}_{\text{lf}}$ to gets the output of each fusion unit.

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After the FFM, the output feature maps of the last fusion unit will be sent into a GAP-FC-SoftMax block to get the final classification map of coastal wetland.

D. Loss Function

In this section, the loss function of MsFE-IFN in training stage is proposed for the HSI and LiDAR coastal wetland classification task. Besides the proposed the orthogonal loss L_{ort} and the JKL loss L_{JKL}, the cross entropy L_{cs} is also combined to jointly optimize the MsFE-IFN. In training stage, all the learnable parameters of MsFE-IFN are initialized and optimized in an end-to-end manner by the sum of these three losses L_{all}. The L_{all} can be calculated as follows:

$$L_{\text{cs}} = \sum_{i=1}^{N} y_i \cdot -\log(p_i)$$
$$L_{\text{all}} = L_{\text{cs}} + \lambda_1 L_{\text{ort}} + \lambda_2 L_{\text{JKL}} \tag{14}$$

where y_i is the one-hot embedding of the ground-truth label, which is the required output. p_i is the prediction probability distribution of each class, which is obtained from the output of MsFE-IFN. N is the number of samples in training stage. The λ_1 and λ_2 are the balance coefficients to control the relative weight of the different loss terms.

IV. Experimental Results and Analysis

A. Experimental Setup

All the experiments are implemented on the platform with NVIDIA GeForce RTX 3090 GPU (24 GB GDDR6X), Intel Core i9-10900K CPU, and 128-GB RAM. All the programs are compiled by the PyTorch 1.10.0 with Python 3.7.11. The Adam algorithm is adopted to optimize all the learnable parameters of MsFE-IFN. In training, the number of epochs is 50, the batch size is set to 128, and the learning rate is 0.00001. In test stage, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (K) are applied to evaluate the classification performance.

B. Sensitivity Analysis

The sensitivity analysis aims to explore the optimal parameters of the model, obtaining the best performance in coastal wetland classification. In this section, how the different patch sizes and hyperparameter affect the MsFE-IFN's classification performance has been evaluated in three coastal wetland datasets. The detailed analysis is as follows.

1) Effect of Different Patch Sizes: The input HSI and LiDAR data of MsFE-IFN are the patch composed of the central pixel and its neighborhood. Different patch sizes carry different spatial scale ground object information. The patch size is a basic parameter affecting the classification performance. Fig. 4 shows the OA in different patch sizes (7×7, 9×9, 11×11, 13×13, and 15×15) with three datasets. It can be find that the classification accuracy shows a trend of first increasing and then decreasing in all datasets. The inflection point for accuracy improvement occurs when the patch size is 11×11. This phenomenon proves that with the increase of patch size, the classification performance of the model will be improved. However, excessive patch size will not bring accuracy enhancement and even reduce classification performance. This is because the large patch size has a large neighborhood, and it is easier to have ground objects that are inconsistent with the central pixel, resulting in redundancy of spatial information.

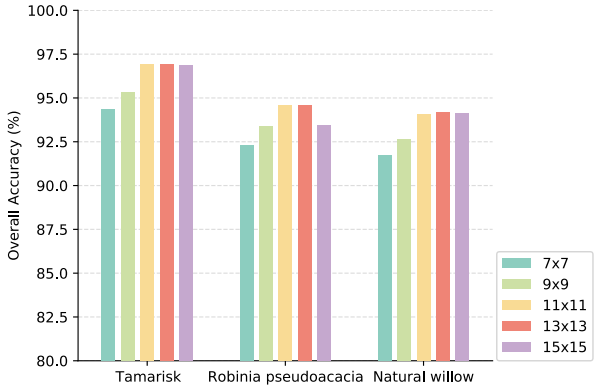


Fig. 4. Classification accuracy on different patch sizes.

In Tamarisk and Natural willow datasets, when the patch size increases from 11×11 to 15×15, classification accuracy tends to be stable or slightly decreased. While in Robinia pseudoacacia dataset, the classification accuracy is stable when the patch size increases from 11×11 to 13×13 and decreases significantly when the patch size changes from 13×13 to 15×15. This can be summarized to the following reasons.

The landscapes of Tamarisk and Natural willow datasets are natural landscapes. The same vegetation can be clumped, and the diameter of the distribution area is more than 6 m. The side length of one pixel is 0.5 m, so the 11×11, 13×13, and 15×15 size patches mostly cover the same class as the center pixel, which does not introduce too much redundant information. However, the landscapes of Robinia pseudoacacia dataset are the artificial landscape, and there is a lot of human activity nearby. The neighborhood of 6–7-m diameter patch covers many classes that are not the same as the center pixel. It will reduce the discrimination ability of features after aggregated neighborhood information.

In summary, the increase in patch size can really improve classification performance, but there is an upper limit. It is important to note that the larger the patch, the greater the computational overhead. Hence, to balance classification performance and computational overhead, the patch size is set to 11×11 on three datasets.

2) Effect of Different Hyperparameters in Loss Function: Appropriate control of the hyperparameters of the loss function can improve the performance of multisource information fusion and enhance the coastal wetland classification accuracy. To evaluate the influence of the hyperparameters λ_1 and λ_2 in loss function, Fig. 5 reports the results of quantitative comparisons in three datasets. It can be confirmed that the best classification performance is achieved when λ_1 is set to 0.3 and λ_2 is set to 0.001, respectively.

By increasing λ_1, the LiDAR data can learn more diverse informative feature representations. Also, the distribution discrepancy between HSI and LiDAR is able to be pulled

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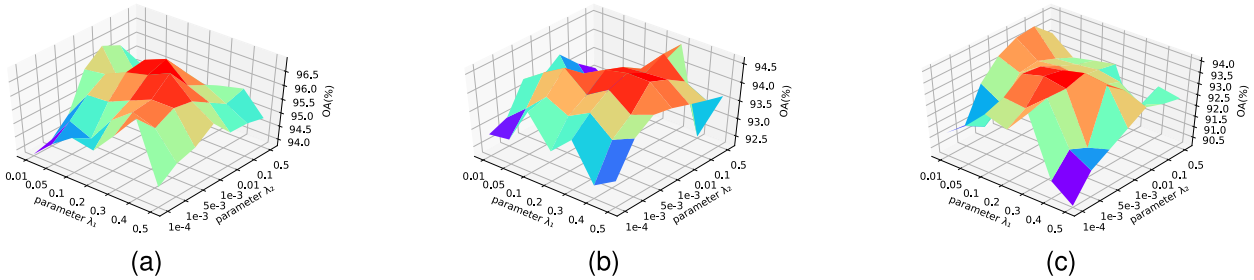


Fig. 5. Classification accuracy on different hyperparameters in loss function: (a) Tamarisk dataset, (b) Robinia pseudoacacia dataset, and (c) Natural willow dataset.

TABLE IV
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON ORTHOGONAL LOSS

	Tamarisk			Robinia pseudoacacia			Natural willow		
	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
Without orthogonal loss	94.78	96.25	94.46	93.38	94.57	92.08	91.90	93.46	91.82
Generating 100 dimensions	95.40	96.77	94.70	92.89	94.14	91.94	92.42	94.27	91.68
With orthogonal loss	**96.91**	**97.20**	**96.08**	**94.61**	**95.64**	**93.54**	**94.06**	**95.93**	**93.02**

close by increasing λ_2. Therefore, the classification accuracy is first improved when λ_1 and λ_2 are increased. However, further increasing λ_1 and λ_2 would force the network put more efforts on minimizing the orthogonal loss and JKL loss, which may cause the network to pay less attention to the classification task and reduce the classification accuracy. As shown in Fig. 5, the classification accuracy of the MsFE-IFN decreases significantly when λ_1 and λ_2 are greater than 0.3 and 0.001, respectively. This phenomenon proves that the classification performance is sensitive to hyperparameters λ_1 and λ_2. Experiments on three datasets also show that using hyperparameters to control the weights of cross-entropy loss, orthogonal loss, and JKL loss can obtain the best classification results.

C. Ablation Analysis

To evaluate the contribution of vital components in the proposed MsFE-IFN, in this section, some ablation studies are conducted on different configurations with three datasets. The detailed analysis is as follows.

1) Ablation Analysis of Orthogonal Loss: The ablation experiment of orthogonal loss is first conducted, which aims to explore the advantage of orthogonal loss in LiDAR features expansion. The LiDAR features expansion is to get the d dimensions' feature. Hence, two comparison approaches are adopted: 1) directly splicing $\boldsymbol{F}^1_{\text{lidar}}$ and $\boldsymbol{F}^2_{\text{lidar}}$ into d dimensions' feature without using orthogonal loss and 2) directly generating one d dimensions' feature. The classification results of orthogonal loss ablation experiments are listed in Table IV. It can be found that the use of orthogonal loss gets OA 96.91%, 94.61%, and 94.06% on Tamarisk, Robinia pseudoacacia, and Natural willow datasets, respectively, which provides approximately 1%–2% improvements for the other two approached. It can be attributed to the following reasons.

1) The two orthogonal n dimensions' features are relatively independent, which can capture the feature distribution and data spatial structure in different directions, and the fused feature has strong feature representation capability and exhibits robustness.
2) The two n dimensions' features without orthogonal loss constraints have strong correlation, and the captured information is overlapping and incomplete, which leads to the lack of diversity of fused feature.
3) Directly generating one d dimensions' feature is to map LiDAR data to a high-dimensional space, and the features become sparse, which may lead to dimensional disasters.

Therefore, the experimental results prove the effectiveness of the orthogonal loss.

2) Ablation Analysis of JKL Loss: The contribution of JKL loss is further studied. Three different comparison loss functions are applied to deal with the distribution discrepancy between HSI and LiDAR. They are cosine distance loss, Euclidean distance loss, and Jensen-Shannon (JS) divergence loss. Table V reports the classification results of these loss functions. Taking the Tamarisk dataset as an example, the OA of without using any distribution distance loss function is only 92.38%, which is much lower than other approaches. It can be verified that using the distribution distances loss function to reduce the distribution difference between HSI and LiDAR is beneficial to improve the classification accuracy. The classification performance of using JKL loss yields K 96.08, about 1.8, 2.0, and 0.9 higher than that in using cosine distance loss, Euclidean distance loss, and JS loss. The same phenomenon occurred in the other two datasets. This can be summarized to the following reasons.

1) The cosine distance loss ignores the magnitude of the vector, which may cause the distribution difference between the HSI and LiDAR in similar directions cannot be effectively reduced.
2) $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ are high-dimensional features, and the Euclidean distance loss is easy to distribute by noise in high-dimensional space and cannot effectively capture the distribution difference.

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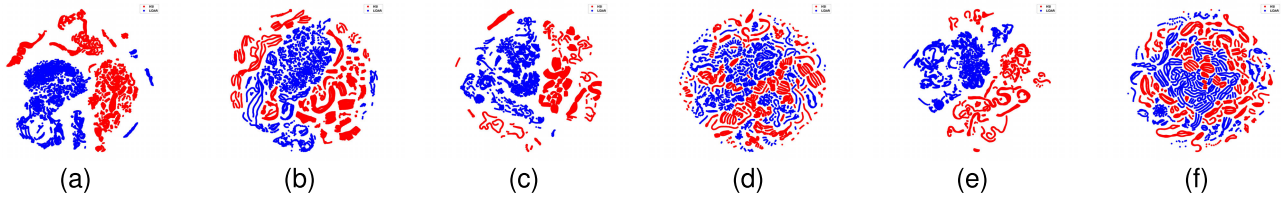


Fig. 6. t-SNE of the performance of HSI-LiDAR feature alignment: (a) original features distribution of Tamarisk dataset, (b) aligned features distribution of Tamarisk dataset, (c) original features distribution of Robinia pseudoacacia dataset, (d) aligned features distribution of Robinia pseudoacacia dataset, (e) original features distribution of Natural willow dataset, and (f) aligned features distribution of Natural willow dataset.

TABLE V
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON JKL LOSS

	Tamarisk			Robinia pseudoacacia			Natural willow		
	OA(%)	AA(%)	K((×100)	OA(%)	AA(%)	K((×100)	OA(%)	AA(%)	K((×100)
Without distribution distances loss	92.38	93.97	92.92	92.10	93.39	92.07	90.21	92.07	90.04
With cosine distance loss	94.80	94.24	94.20	93.93	94.40	92.73	92.37	93.10	91.22
With euclidean distance loss	94.46	94.34	94.06	94.18	95.24	93.13	92.56	92.10	91.78
With JS loss	95.88	96.38	95.12	94.09	94.69	92.92	93.13	94.21	92.53
With JKL loss	**96.91**	**97.20**	**96.08**	**94.61**	**95.64**	**93.54**	**94.06**	**95.93**	**93.02**

TABLE VI
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON CROSS-MODALITY FEATURE INTERACTION

	Tamarisk			Robinia pseudoacacia			Natural willow		
	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
Without cross-modality feature interaction	91.62	92.73	92.47	92.89	92.53	91.68	91.77	93.30	90.48
Only spatial-wise feature interaction	94.40	95.20	94.45	93.82	94.16	92.60	92.55	94.21	92.18
Only channel-wise feature interaction	94.32	95.05	94.35	93.91	94.80	92.70	92.43	94.00	92.55
With cross-modality feature interaction	**96.91**	**97.20**	**96.08**	**94.61**	**95.64**	**93.54**	**94.06**	**95.93**	**93.02**

3) Compared with JKL loss, JS loss has symmetry, which may result in the distribution of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$ influencing each other. JKL loss calculates the KL divergence for $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$, respectively, which comprehensively considers the feature distribution of $\boldsymbol{F}_H$ and $\boldsymbol{F}_L$, thereby reducing the distribution difference.

According to the experimental results, the JKL loss could effectively reduce the distribution difference between HSI and LiDAR. Fig. 6 shows the t-distributed stochastic neighbor embedding (t-SNE) of the performance of HSI-LiDAR feature alignment, in which the aligned features have more consistent distribution.

3) Ablation Analysis of Cross-Modality Feature Interaction: To evaluate the effectiveness of cross-modality feature interaction, the ablation experiments of the proposed MsFE-IFN without cross-modality feature interaction, only with spatialwise feature interaction, and only with channelwise feature interaction are conducted. As listed in Table VI, there is obvious performance degradation when the MsFE-IFN is deployed without cross-modality feature interaction. Taking the Natural willow dataset as an instance, the OA of "without cross-modality feature interaction" is only about 91.77%, which is the lowest accuracy in ablation experiments. The OAs of "only spatialwise feature interaction" and "only channelwise feature interaction" are 92.55% and 92.43%, respectively. By contrast, the OA of "with cross-modality feature interaction" yields 94.06%, which is higher about 2% than that in other feature interaction methods. It is clear that only obtaining the spectral–elevation or the geospatial information cannot fully utilize the advantages of features fused by HSI and LiDAR. Based on the experiments results provided in Table VI, it can be verified that the interrelation of spectral–elevation–geospatial information captured by cross-modality feature interaction is effective in improving the performance of coastal wetlands classification.

4) Ablation Analysis of Sequence-to-Sequence Fusion: The sequence-to-sequence fusion is the first stage to fuse the HSI and LiDAR features into a single feature. To verify the contribution of the sequence-to-sequence fusion, the ablation experiment is conducted, and the experiment results are reported in Table VII. In Table VII, when the MsFE-IFN is applied without sequence-to-sequence fusion, the complementary information in two modalities cannot be extracted, resulting the classification performance of "without sequence-to-sequence fusion" clearly lower than that of "with sequence-to-sequence fusion." According to Table VII, the OA of "with sequence-to-sequence fusion" is higher about 1.6%, 1.1%, and 1.4% than that of "without sequence-to-sequence fusion" in Tamarisk, Robinia pseudoacacia, and Natural willow datasets, respectively. The reason for this is that the cross attention consisting in sequence-to-sequence fusion is able to find the most valuable information about this modality in another modality and integrating the complementary information. The same phenomenon is also appeared in AA and K, showing the effectiveness of sequence-to-sequence fusion for final classification.

5) Ablation Analysis of Low-Frequency Residual Enrichment: The low-frequency residual enrichment is the second

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TABLE VII
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON SEQUENCE-TO-SEQUENCE FUSION

	Tamarisk			Robinia pseudoacacia			Natural willow		
	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
Without sequence-to-sequence fusion	95.33	96.58	94.36	93.54	94.02	92.28	92.66	93.46	92.55
With sequence-to-sequence fusion	**96.91**	**97.20**	**96.08**	**94.61**	**95.64**	**93.54**	**94.06**	**95.93**	**93.02**

TABLE VIII
CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON LOW-FREQUENCY RESIDUAL ENRICHMENT

	Tamarisk			Robinia pseudoacacia			Natural willow		
	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)	OA(%)	AA(%)	K(×100)
Without low-frequency residual enrichment	94.96	96.09	94.03	92.25	93.33	92.18	92.88	93.79	92.27
With low-frequency residual enrichment	**96.91**	**97.20**	**96.08**	**94.61**	**95.64**	**93.54**	**94.06**	**95.93**	**93.02**

stage to fuse the HSI and LiDAR features into a single feature. The low-frequency residual enrichment aims to recapture the low-frequency information, which has intraclass-invariant characteristic and easily losing during the convolution process. Table VIII lists the ablation experiment results of low-frequency residual enrichment. It can be shown that the "with low-frequency residual enrichment" achieves better performance on both three datasets. Taking Robinia pseudoacacia dataset as an example, the "with low-frequency residual enrichment" achieves higher OA, AA, and K by 2.4%, 2.3%, and 1.4 than that of "without low-frequency residual enrichment." The excellent performance of low-frequency residual enrichment has the following reasons. Convolution operations introduce noise when they extract high-frequency feature with fine-grained information. Low-frequency residual can help the network to capture the intraclass-invariant characteristic, reducing the influence of noise, which is able to improve the classification performance. Accordingly, the low-frequency residual enrichment mechanism is effective for final classification.

D. Classification Performance

1) Classification Performance Compared With Other Methods: To validate the effectiveness of the proposed MsFE-IFN, different competitive methods are employed for comparison, which include three classification models originally developed for HSI and eight HSI-LiDAR classification models. The HSI-based methods include the support vector machine (SVM), the context deeper CNN (CDCNN) [43], and the 3DCNN [44]. Besides, the HSI-LiDAR classification models are the multiscale network with self-calibrated CNN (MSNetSC) [37], the attention-based multimodal fusion network (FusAtNet) [45], the morphological CNN (MorphCNN) [46], the two-branch CNN (TBCNN) [47], the spatial–spectral cross-modal enhancement network (S2ENet) [48], the coupled CNNs [35], multimodal fusion transformer (MFT) network [49], and hierarchical CNN with transformer (HCT) network [39]. For a fairer comparison, the HSI and LiDAR are concatenated into a cube as the input of SVM, CDCNN, and 3DCNN. All the models are deployed following the configurations provided in the corresponding paper, and the parameters are adjusted to the optimum. For a fair comparison, the patch size of the comparative methods is also set to 11 × 11. Tables IX–XI report the performance comparative experiments results of different methods in three datasets. It can be seen that the proposed MsFE-IFN is superior to other methods.

Taking the Tamarisk dataset as an example, the best OA, AA, and K of SVM are 88.71%, 88.76%, and 85.83, respectively, which is obviously lower than that of other methods. Due to the other methods owning to the DL-based methods, this verifies the advantage of DL in the coastal wetlands classification. Except MSNetSC, the OA of other HSI-LiDAR classification models is higher than CDCNN and 3DCNN, which shows that the multisource data fusion classification method is more suitable for HSI and LiDAR joint classification task. In addition, the MsFE-IFN yields OA 96.91%, about 6.5% better than that of MSNetSC (i.e., 90.49%), 3.6% better than that of FusAtNet (i.e., 93.35%), 2.5% better than that of MorphCNN (i.e., 94.42%), 2.3% better than that of TBCNN (i.e., 94.62%), 1.9% better than that of S2ENet (i.e., 95.00%), and 1% better than that of CoupledCNNs (i.e., 95.93%), MFT (i.e., 95.53%), and HCT (i.e., 96.08%). In particular, MsFE-IFN consistently leads in classification accuracy across all classes. The excellent performance is due to the distribution alignment of the MsFE-IFN, which reduces the HSI and LiDAR feature distribution differences. Also, the feature interaction also improves the discriminability of fusion features. Fig. 7 further illustrates the classification performance of all the methods on the Tamarisk dataset. Compared with other methods, MsFE-IFN provides a smoother noise-free visualization results.

Table X shows the classification classification performance of different methods on the Robinia pseudoacacia dataset. The classification accuracy of all methods is lower than that of Tamarisk dataset. The reason is that Robinia pseudoacacia dataset has more classes number and more similar observation objects. The proposed MsFE-IFN still has the best performance among all comparison methods in OA and K. However, The AA of MsFE-IFN is 95.64%, about 0.02% lower than that of CoupledCNNs (i.e., 95.66). This is mainly influenced by the classification accuracy of "Robinia pseudoacacia," "Salix matsudana," and "Withered Robinia pseudoacacia." These three classes are mixed growth and similar in elevation; hence, the classification performance is limited. For the classification accuracy of other classes, the proposed MsFE-IFN is superior to other methods, and the Tamarix chinensis even

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TABLE IX

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE TAMARISK DATASET

Class	Method											
	SVM	CDCNN	3DCNN	MSNetSC	FusAtNet	MorphCNN	TBCNN	S2ENet	CoupledCNNs	MFT	HCT	MsFE-IFN
Small tidal creek	97.47	91.55	94.07	95.30	94.26	97.28	97.47	**99.50**	98.82	98.82	98.98	99.27
Phragmites australis	90.90	96.27	92.62	91.57	97.97	95.46	97.24	**98.42**	96.67	97.02	96.01	98.18
Desiccated Spartina alterniflora	97.54	96.41	96.83	95.03	95.42	98.24	96.39	**98.84**	97.96	98.31	98.30	98.23
Abandoned oil platform	88.67	90.00	88.67	82.67	86.67	90.67	95.33	100.0	100.0	99.33	100.0	**100.0**
Tamarix chinensis	67.15	72.49	74.47	79.93	87.54	85.99	92.13	80.87	92.47	91.69	91.19	**94.85**
Suaeda salsa	76.50	83.62	84.89	81.21	84.66	88.22	87.61	**94.32**	90.26	86.61	91.48	91.34
Large tidal creek	99.10	98.47	98.80	96.41	97.78	98.84	98.85	99.22	99.33	98.87	98.75	**99.45**
TC and SS mixed	86.62	84.71	71.81	93.59	98.27	92.73	79.31	**99.71**	87.87	96.22	98.07	93.89
Roads	94.91	97.25	98.51	95.21	97.77	98.10	96.92	98.40	99.63	98.36	**99.74**	99.63
OA(%)	88.71±2.33	90.30±1.25	90.54±1.70	90.49±1.85	93.35±1.19	94.42±1.82	94.62±1.79	95.00±1.22	95.93±1.85	95.53±1.46	96.08±2.62	**96.91±1.23**
AA(%)	88.76±1.46	90.09±2.63	88.96±1.29	90.10±1.84	93.37±0.74	93.95±2.09	93.47±1.41	96.59±1.37	95.90±1.04	96.14±1.25	96.95±1.35	**97.20±2.01**
K(×100)	85.83±2.15	87.83±1.63	88.11±0.98	88.06±1.90	91.64±1.20	92.95±3.12	93.20±1.68	93.68±1.60	94.86±1.94	94.35±2.41	95.05±2.33	**96.08±1.45**

TABLE X

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE ROBINIA PSEUDOACACIA DATASET

Class	Method											
	SVM	CDCNN	3DCNN	MSNetSC	FusAtNet	MorphCNN	TBCNN	S2ENet	CoupledCNNs	MFT	HCT	MsFE-IFN
Phragmites australis	73.28	72.79	79.37	80.77	85.89	87.72	86.67	83.80	82.55	76.73	**90.37**	89.17
Robinia pseudoacacia	74.60	77.29	78.25	67.09	79.77	80.10	70.97	79.20	**94.49**	72.69	86.67	90.58
Water	97.93	98.14	96.43	98.14	98.64	98.57	97.32	99.57	**99.83**	96.38	98.01	99.27
Salix matsudana	74.04	73.94	69.29	72.70	81.74	82.89	80.58	**92.47**	91.35	78.13	81.52	89.73
Tamarix chinensis	95.22	91.93	94.92	97.91	99.55	99.10	99.25	99.85	99.40	96.41	97.01	**100.0**
Rice paddy	95.60	91.59	93.37	95.62	97.18	97.26	97.38	98.16	**99.47**	96.02	93.70	99.02
Highway	98.66	99.06	98.64	99.01	99.46	99.62	99.67	99.71	99.77	**99.87**	99.82	99.82
Abandoned farmland	96.12	95.44	95.41	97.69	98.05	97.21	99.00	98.67	98.86	97.60	96.82	**99.60**
Firebreak ditch	97.72	97.07	95.60	99.86	99.96	99.95	99.98	99.62	99.92	98.29	97.94	**99.99**
Dirt road	97.03	98.20	90.49	98.80	98.49	98.95	99.78	98.19	98.46	97.14	99.34	**99.85**
Withered Robinia pseudoacacia	82.13	82.56	81.24	78.86	83.05	90.64	85.45	**94.75**	88.20	90.60	82.93	84.97
OA(%)	85.95±2.41	85.82±0.95	86.53±1.29	86.20±1.88	90.73±1.98	91.33±1.47	89.06±1.51	91.79±1.48	94.29±2.53	86.64±1.24	92.12±2.49	**94.61**±0.84
AA(%)	89.30±4.52	88.91±1.29	88.45±2.63	89.68±1.06	92.89±1.37	93.82±1.15	92.37±1.58	94.91±1.89	**95.66**±2.45	90.90±1.45	93.10±2.01	95.64±0.55
K(×100)	83.29±3.26	83.14±1.98	83.96±1.60	83.61±1.98	88.94±1.52	89.64±2.36	86.98±1.93	90.21±1.49	93.15±3.02	84.17±2.51	90.56±3.12	**93.54**±1.14

TABLE XI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON THE NATURAL WILLOW DATASET

Class	Method											
	SVM	CDCNN	3DCNN	MSNetSC	FusAtNet	MorphCNN	TBCNN	S2ENet	CoupledCNNs	MFT	HCT	MsFE-IFN
Salix matsudana	73.17	71.78	78.26	76.89	80.97	84.67	81.71	81.25	86.77	75.85	80.99	**88.95**
Senescent Phragmites australis	92.24	93.82	93.47	95.35	92.82	95.40	90.92	**96.57**	96.84	92.23	93.76	95.59
Populus	87.21	88.70	91.10	81.59	91.48	92.58	94.89	95.68	95.53	93.31	**96.75**	88.12
Tamarix chinensis	83.14	59.70	58.10	86.00	88.97	85.08	89.17	**91.98**	89.45	88.57	80.97	88.31
Yellow River diversion canal	91.26	93.88	97.10	95.13	98.53	97.22	97.90	**99.43**	97.66	97.10	98.34	98.07
Dirt road 1	95.00	98.28	96.47	96.19	99.13	96.83	98.56	99.59	97.48	98.00	98.99	**99.68**
Dirt road 2	87.91	83.00	88.06	92.34	95.17	93.54	98.07	96.77	99.22	86.34	98.01	**99.96**
Highway	96.86	97.55	98.13	96.74	97.30	98.98	99.00	99.59	**99.93**	99.00	99.84	99.80
Farmland	94.11	94.37	96.08	97.63	99.67	99.18	99.84	**100.0**	98.16	98.35	99.93	99.97
Yellow River riverbed	92.00	96.18	93.08	95.85	**97.94**	94.78	97.01	95.11	97.15	92.24	97.02	97.37
Pool	99.89	99.50	99.83	99.02	99.30	99.94	99.43	99.97	99.92	99.96	99.11	**99.99**
Phragmites australis	90.65	83.39	91.66	88.80	95.49	92.75	**96.21**	95.02	92.46	92.00	95.20	95.37
OA(%)	87.28±4.56	84.53±1.26	87.04±1.09	89.35±2.79	91.96±1.38	91.97±2.31	91.84±4.20	92.96±1.54	93.58±1.65	89.34±3.21	91.42±1.36	**94.06**±1.16
AA(%)	90.29±2.17	88.35±1.31	90.11±1.32	91.79±1.84	94.73±2.05	94.24±1.25	95.26±3.65	95.91±1.49	95.88±1.19	92.75±2.15	94.91±1.05	**95.93**±1.18
K(×100)	85.14±4.64	81.90±1.22	84.81±1.61	87.53±2.18	90.58±4.46	90.58±2.36	90.45±4.27	91.77±2.69	92.46±2.22	87.55±4.12	89.95±1.88	**93.02**±1.23

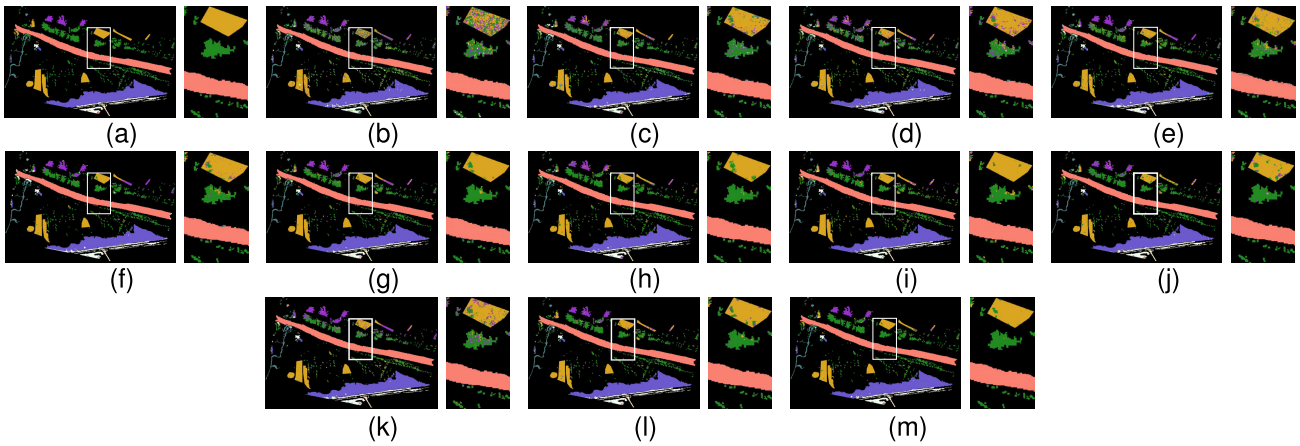


Fig. 7. Ground-truth map and classification maps of Tamarisk dataset: (a) ground-truth map and (b)–(m) classification maps of SVM, CDCNN, 3DCNN, MSNetSC, FusAt, MorphCNN, TBCNN, S2ENet, coupled CNNs, MFT, HCT, and MsFE-IFN.

achieves complete accuracy, which verify the effectiveness of the MsFE-IFN. Fig. 8 demonstrates the visual maps for qualitative evaluation of all methods on Robinia pseudoacacia dataset. It can be seen that the proposed MsFE-IFN has a low false detection.

The classification performance of different methods on the Natural willow dataset is listed in Table XI. Owning to the mass mixed growth of Tamarix chinensis and Salix matsudana, the Natural willow dataset is the hardest to classify in three datasets. Even so, the proposed MsFE-IFN obtains the best

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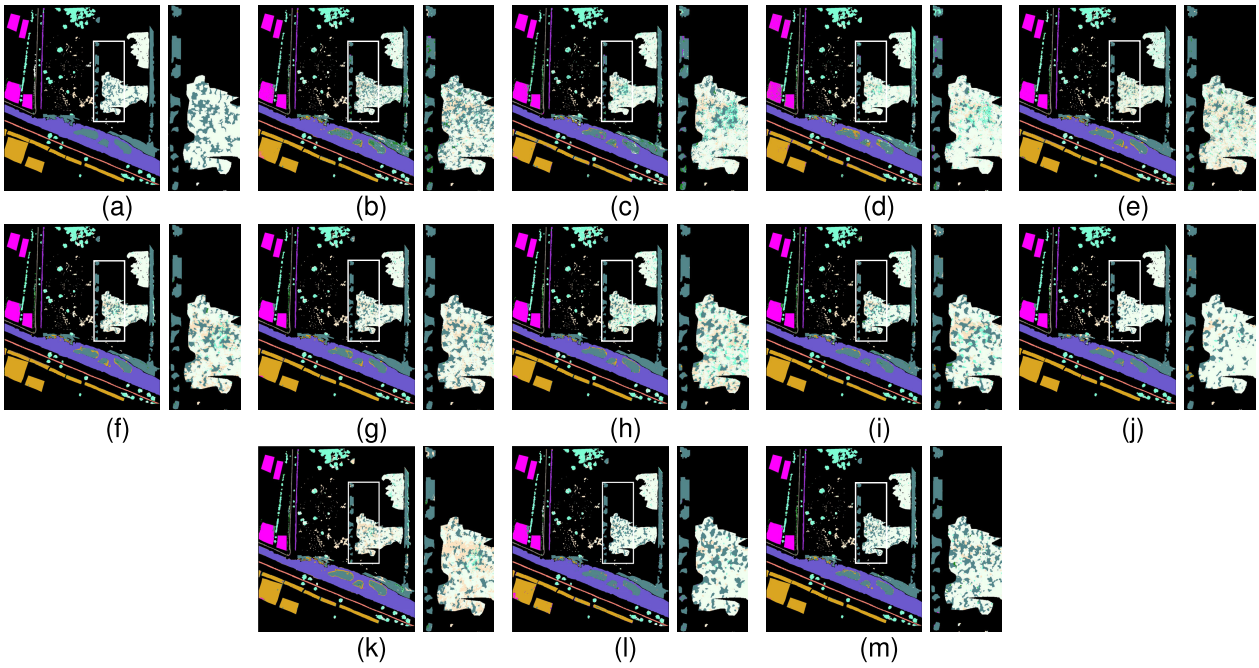


Fig. 8. Ground-truth map and classification maps of Robinia pseudoacacia dataset: (a) ground-truth map and (b)–(m) classification maps of SVM, CDCNN, 3DCNN, MSNetSC, FusAt, MorphCNN, TBCNN, S2ENet, coupled CNNs, MFT, HCT, and MsFE-IFN.

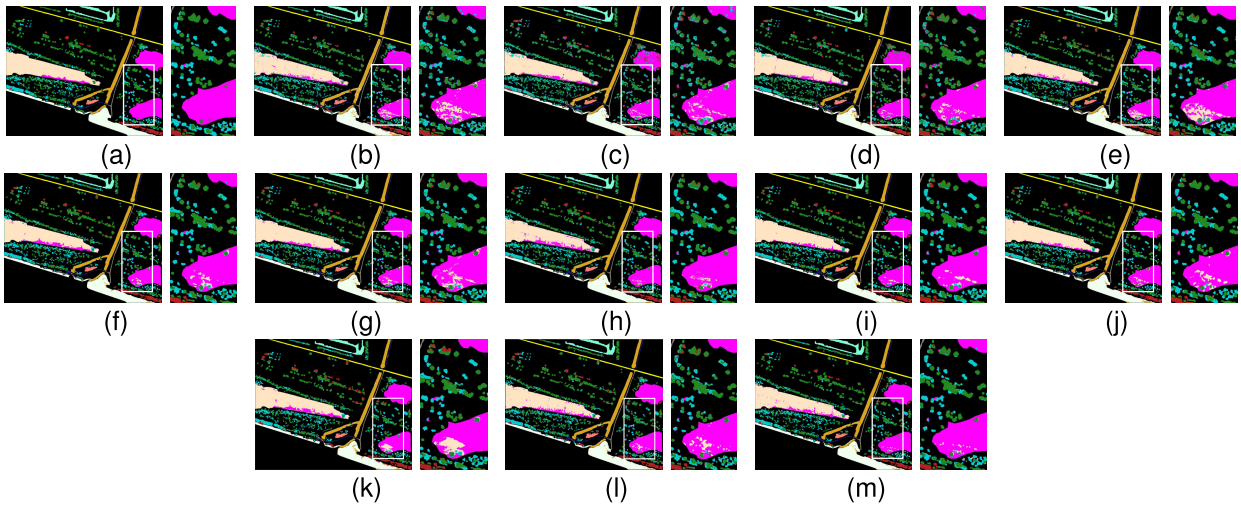


Fig. 9. Ground-truth map and classification maps of Natural willow dataset: (a) ground-truth map and (b)–(m) classification maps of SVM, CDCNN, 3DCNN, MSNetSC, FusAt, MorphCNN, TBCNN, S2ENet, coupled CNNs, MFT, HCT, and MsFE-IFN.

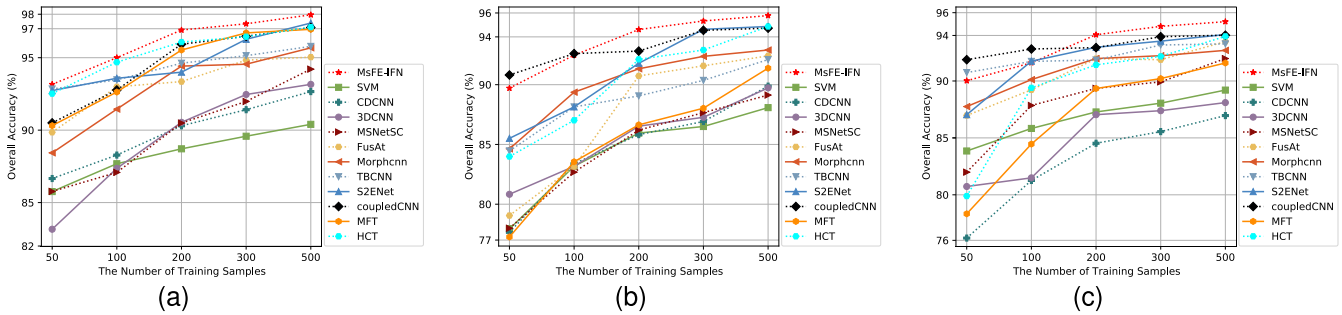


Fig. 10. Classification performance with different numbers of training samples: (a) Tamarisk dataset, (b) Robinia pseudoacacia dataset, and (c) Natural willow dataset.

classification performance among all comparison methods. For instance, MsFE-IFN provides approximately 6.8%, 9.5%, 7.0%, 3.7%, 2.1%, 2.1%, 2.2%, 1.1%, 0.5%, 4.7%, and 2.6% improvements for SVM, CDNN, 3DCNN, MSNetSC,

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TABLE XII
CALCULATION AND SPACE COMPLEXITY OF DIFFERENT METHODS ON THE TAMARISK DATASET

Complexity	Method										
	CDCNN	3DCNN	MSNetSC	FusAtNet	MorphCNN	TBCNN	S2ENet	CoupledCNNs	MFT	HCT	MsFE-IFN
FLOPs	35.679M	83.763M	8.226M	4.007G	47.381M	112.465M	30.257M	2.295M	22.081M	31.537M	520.006M
Parameters	305.755K	293.024K	46.683K	37.942M	208.937K	1.770M	249.938K	102.459K	298.281K	871.817K	15.546M

TABLE XIII
TIME COMPLEXITY OF DIFFERENT METHODS ON THE ROBINIA PSEUDOACACIA AND NATURAL WILLOW DATASETS

Dataset	Time	Method											
		SVM	CDCNN	3DCNN	MSNetSC	FusAtNet	MorphCNN	TBCNN	S2ENet	CoupledCNNs	MFT	HCT	MsFE-IFN
Robinia pseudoacacia	Training(s)	168.214	397.928	264.877	1365.305	163.994	166.739	137.583	353.746	320.841	186.703	65.722	60.644
	Testting(s)	54.516	70.970	41.008	170.425	214.919	28.266	24.762	20.667	73.022	28.578s	22.374	45.188
Natural willow	Training(s)	195.794	422.208	335.853	1492.536	177.851	168.791	144.238	371.806	325.577	200.240	69.036	64.404
	Testting(s)	53.639	66.944	38.638	154.423	172.024	25.910	23.371	18.966	77.211	25.790	21.682	42.307

MorphCNN, FusAt, TBCNN, S2ENet, coupled CNN, MFT, and HCT, respectively, in OA. The same phenomenon can be found in AA and K. Moreover, MsFE-IFN shows stable classification performance in each class accuracy, with all classes accuracy exceeding 88% and most classes accuracy exceeding 95%. It demonstrates the robustness of the MsFE-IFN in coastal wetland classification, verifying the advantage of distribution alignment and feature interaction. The visual comparison on Natural willow dataset is further shown in Fig. 9, and the classification map of MsFE-IFN is also significantly better than other classifiers.

2) Classification Performance With Different Numbers of Training Samples: The number of training samples plays a vital role in the final classification. Fig. 10 illustrates the classification performance of all comparison methods with different training samples. In particular, the number of training samples in each class is set to 50, 100, 200, 300, and 500 for three datasets. In general, the more the training samples, the higher the classification accuracy of the model. Fig. 10 shows that the classification performance of all methods is positively correlated with the number of training samples. Compared with other methods, the proposed MsFE-IFN does not have no obvious advantage in classification accuracy, when the number of training samples is 50 or 100, but it is relatively flat. However, when the number of training samples is 200, 300, or 500, the classification performance of MsFE-IFN is significantly better than other models. The reason may be that the distribution alignment of HSI and LiDAR requires a certain number of sample; otherwise, they cannot be mapped to the unified space. Even so, 200–500 samples are not difficult to obtain for coastal wetlands. Therefore, the classification results indicate the effectiveness of MsFE-IFN for coastal wetland classification.

E. Complexity Analysis

Model complexity is another important indicator of the model performance. In this section, the floating-point operations (FLOPs), the number of parameters, and the time costing are applied to evaluate the calculation complexity, space complexity, and time complexity. Taking the Tamarisk dataset as an example, Table XII reports the FLOPs and the number of parameters in different comparison methods. It can be observed that the MsFE-IFN has the second-most FLOPs and parameters, which is less than FusAtNet. The reason why MsFE-IFN has the many FLOPs and parameters is that the feature interaction requires lots of learnable parameters. Fortunately, at the cost of calculation and space, MsFE-IFN achieves excellent classification performance. Taking the Robinia pseudoacacia and Natural willow dataset as examples, Table XIII lists the training–testing time costing in different comparison methods. In Table XIII, the proposed MsFE-IFN has the shortest training time on two datasets, only about 60 s and the testing time only about 40 s. This provides a possibility for the rapid deployment of MsFE-IFN in coastal wetlands.

V. CONCLUSION

In this work, we propose an MsFE-IFN for the coastal wetland classification with HSI and LiDAR. The MsFE-IFN aims to alleviate the data distribution differences of HSI and LiDAR and better fuse the HSI and LiDAR for collaborative classification. Specifically, the DFE-DAM is presented to generate the diverse LiDAR features and align the distribution of HSI features and diverse LiDAR features. Moreover, the CMFIM is designed to capture the interrelation of spectral–elevation–geospatial information, enhancing the discriminability of features. Finally, the two-stage FFM is developed to establish the sequence-to-sequence fusion and retain the low-frequency feature. Both modules are integrated into a unified framework, which can train quickly and deploy efficiently. Extensive experiments conducted on three coastal wetland datasets of Yellow River Delta reveal the effectiveness of MsFE-IFN, which can outperform other state-of-the-art models in coastal wetland classification task. It is worth noting that the classification performance of MsFE-IFN is not satisfying when identifying mixed vegetation with similar elevation. One of the future works is expected to deal with this problem.

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